

HIROSHIMA, Japan

WMO: 477650

Lat: 34.3983N

Lon: 132.4628E

Elev: 54

StdP: 100.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	0.1	-8.3	1.9	6.0	-6.7	2.1	5.8	9.6	7.1	8.6	7.0	3.5	10	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	34.0	25.1	33.0	25.1	32.0	24.7	26.2	31.4	25.7	30.7	25.4	30.2	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.0	28.8	24.4	19.5	28.5	24.0	19.0	28.3	81.8	31.6	79.9	30.9	78.2	30.3	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.0	7.9	7.0	DB	-2.6	35.7	1.1	0.9	-3.4	36.4	-4.0	36.9	-4.6	37.4	-5.4	38.1
(5)				WB	-3.7	26.9	1.3	0.5	-4.6	27.2	-5.3	27.6	-6.0	27.9	-6.9	28.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	5.6	6.4	9.6	14.8	19.7	23.3	27.5	28.8	25.0	19.2	13.1	7.8
	DBStd	8.39	2.23	2.65	2.97	3.01	2.18	1.99	2.24	1.77	2.61	2.81	3.07	2.71
	HDD10.0	374	138	105	44	3	0	0	0	0	0	7	77	
	HDD18.3	1628	395	333	272	111	12	0	0	0	22	157	326	
	CDD10.0	2856	1	4	31	146	301	399	542	584	450	287	101	10
	CDD18.3	1068	0	0	0	4	54	149	283	326	201	51	1	0
	CDH23.3	9934	0	0	0	16	230	855	3051	3935	1645	202	0	0
	CDH26.7	3618	0	0	0	1	16	147	1191	1752	494	17	0	0
Wind	WSAvg	3.6	3.5	3.6	3.8	3.7	3.4	3.1	3.2	3.4	3.8	4.1	3.9	3.7
Precipitation	PrecAvg	1593	48	63	113	156	157	259	238	122	181	105	72	44
	PrecMax	2228	137	158	238	277	400	659	512	462	445	388	201	107
	PrecMin	1114	4	11	21	51	40	55	15	6	13	23	2	2
	PrecStd	265	29	34	55	56	77	116	148	102	98	70	49	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.7	20.2	25.4	28.3	30.9	35.0	35.5	33.2	28.5	22.6	16.7
		MCWB	8.5	12.1	12.7	16.2	17.9	23.4	25.3	25.0	24.1	21.8	16.7	11.1
	2%	DB	12.2	14.4	18.3	23.1	26.8	29.4	33.6	34.4	31.8	26.7	21.1	15.2
		MCWB	7.7	9.6	11.9	14.8	17.5	21.9	25.4	25.1	23.7	19.9	15.5	10.6
	5%	DB	11.0	12.7	16.6	21.6	25.7	28.4	32.6	33.5	30.7	25.5	19.7	13.9
		MCWB	6.6	8.1	10.9	14.3	17.2	21.3	25.1	25.1	23.4	18.9	14.7	9.5
	10%	DB	9.8	11.2	15.1	20.2	24.6	27.3	31.6	32.6	29.5	24.3	18.4	12.6
		MCWB	5.8	7.0	10.2	13.6	16.8	20.8	24.8	24.9	22.8	18.2	13.9	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	13.5	15.1	19.1	21.5	24.7	26.6	27.1	25.9	23.5	18.7	13.4
		MCDB	12.9	16.1	17.5	21.6	24.1	28.8	32.4	32.6	30.3	26.7	20.3	14.9
	2%	WB	8.6	10.8	13.5	17.5	20.3	23.9	26.1	26.4	25.2	22.1	17.3	11.5
		MCDB	10.8	13.2	16.1	19.9	23.0	27.4	31.7	31.6	29.3	25.3	19.7	13.9
	5%	WB	7.3	9.1	12.2	16.2	19.4	23.2	25.6	26.0	24.6	20.5	16.0	10.3
		MCDB	10.0	11.6	15.5	19.6	22.5	26.3	30.9	30.9	28.6	23.9	18.5	13.1
	10%	WB	6.3	7.6	10.8	15.2	18.5	22.5	25.2	25.6	24.0	19.2	14.6	9.1
		MCDB	9.2	10.5	14.7	18.8	22.3	25.6	30.1	30.5	28.1	22.8	17.7	12.1
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.7	8.6	8.9	8.5	6.7	6.2	6.9	7.2	8.1	7.9	7.4
		MCDBR	8.9	9.5	10.5	10.5	10.2	8.4	7.6	8.1	8.1	9.1	8.9	8.9
		MCWBR	5.6	6.0	6.3	5.4	4.5	3.4	2.6	2.5	3.0	4.2	5.1	5.5
	5% WB	MCDBR	7.9	7.9	8.5	7.4	6.8	5.5	6.9	6.9	6.5	6.9	7.1	7.7
		MCWBR	5.8	6.0	6.5	4.9	4.2	2.7	2.5	2.5	2.8	4.2	5.1	5.7
Clear-Sky Solar Irradiance	taub	0.412	0.456	0.520	0.549	0.548	0.566	0.566	0.534	0.491	0.445	0.422	0.405	
	taud	2.098	2.009	1.862	1.847	1.909	1.942	1.996	2.085	2.162	2.193	2.159	2.134	
	Ebn at Noon	761	770	753	756	762	746	743	760	774	776	747	742	
	Edh at Noon	126	153	193	205	195	188	177	160	141	126	118	115	
All-Sky Solar Radiation	RadAvg	2.17	2.83	3.87	4.77	5.31	4.63	4.85	5.03	4.04	3.38	2.47	1.91	
	RadStd	0.16	0.24	0.31	0.43	0.53	0.48	0.66	0.49	0.40	0.33	0.20	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-0.60	N/A	-0.37	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page